

An Encounter-Level Neural Network Surrogate for the

SIMON Hybrid Three-Body Gravitational Integrator

Aarush Gupta
UWCSEA Singapore

Abstract

We present an encounter-level neural network surrogate (EncounterNN) that learns the 18-dimensional exit-state residual between the SIMON-noNN leapfrog integrator and the high-accuracy IAS15 reference over a 0.5-year gravitational encounter window in the unrestricted three-body problem. The model, a four-hidden-layer MLP (18→128→128→128→64→18, 44,882 parameters), is trained on 10,870 encounter samples across three timesteps ($dt = 0.06, 0.08, 0.10$ yr) for IC1. At the optimal operating timestep ($dt = 0.08$ yr), the EncounterNN achieves 93.7% exit-state position improvement and 94.0% velocity improvement against IAS15, with +11.9% global final-position improvement over $T = 100$ yr at 2.0x speedup. A controlled comparison against Zone-3 leapfrog fine-stepping confirms EncounterNN is the superior strategy at $dt = 0.08$ on all metrics. We identify the post-encounter Lyapunov horizon as the binding constraint on operating timestep, and derive a quantitative selection criterion: $(T - t_{\text{enc}})/\lambda_L < 3$, where λ_L is the empirically measured Lyapunov time. This criterion correctly identifies $dt = 0.08$ yr as the unique valid operating point among the three candidates tested for IC1.

1. Introduction

The gravitational three-body problem has no general closed-form solution and exhibits sensitive dependence on initial conditions, a hallmark of classical chaos. High-accuracy adaptive integrators such as IAS15 [Rein & Spiegel 2015] provide reliable long-horizon trajectories but at significant computational cost. Leapfrog integrators achieve substantially higher throughput but accumulate integration error that grows with timestep, particularly during close encounters between bodies.

The SIMON (Symplectic Integrator with Machine-learning Online Newtonian correction) hybrid integrator partitions the gravitational force into four distance zones, applying different physics in each zone. Zone 2 ($r < 0.05$ AU) uses exact Newtonian force with adaptive sub-stepping, eliminating the ejection instability demonstrated in earlier work. Zone 3 (0.052 to 0.148 AU) is the focus of this work: it is the region where close-but-not-critical gravitational encounters accumulate trajectory error that cannot be eliminated by temporal refinement alone.

The central contribution of this paper is an encounter-level neural network surrogate that learns the 18-dimensional residual between the coarse leapfrog exit state and the IAS15 reference exit state across a fixed encounter window. Unlike per-step scalar force corrections, the encounter surrogate targets the quantity that directly determines post-encounter trajectory quality. Unlike trajectory replacement approaches, it is a residual correction model that adds a small learned delta to an existing integrator output, preserving physical structure.

This paper is structured as follows. Section 2 presents the SIMON zone architecture and motivates the zone-specific design choices. Section 3 reviews related work. Section 4 describes the EncounterNN methodology

including data generation, model architecture, and deployment. Section 5 defines the evaluation metrics. Section 6 presents results at all three tested timesteps. Section 7 derives the post-encounter Lyapunov criterion that governs optimal timestep selection. Section 8 compares EncounterNN against Zone-3 leapfrog fine-stepping. Section 9 discusses findings and Section 10 lists contributions. Section 11 concludes.

2. SIMON Zone Architecture

SIMON partitions pair separations into four force zones with different physics in each. Figure 1 illustrates the zone structure.



Figure 1: SIMON zone structure. Zone 3 (0.052 to 0.148 AU, blue) is the domain of EncounterNN. Zone 2 ($r < 0.05$ AU) uses exact Newtonian force with adaptive sub-stepping and does not involve neural network correction.

Zone	Separation (AU)	Force Law	Problem Addressed
Zone 1	$r < 3 \times 10^{-4}$	Hard Newtonian fallback	Near-singularity stability
Zone 2	$r < 0.05$	Direct Newtonian + adaptive sub-steps	Temporal resolution at close approach
Zone 3	0.052 to 0.148	Softened Newtonian (learnable)	Force integration accuracy
Zone 4	$r \geq 0.15$	Standard Newtonian	Far-field — exact and fast

Table 1: SIMON zone partitioning and the physics motivation for each.

2.1 Why the NN Is Not Applied to Zone 2

Zone 2 and Zone 3 address fundamentally different integration problems. Zone 2 ($r < 0.05$ AU) is a **temporal resolution problem**: when two bodies approach within 0.05 AU, the gravitational encounter evolves faster than any macro-timestep can resolve. The leapfrog makes one large step across a region where the force changes dramatically. The correct solution is temporal resolution — split the step into sub-steps so the force is evaluated at more points. Adaptive sub-stepping does exactly this and is the mathematically exact, physics-preserving solution requiring no approximation. No machine learning is needed or appropriate.

Zone 3 (0.052 to 0.148 AU) is a **force integration accuracy problem**. The pair is far enough apart that one macro-step resolves the encounter in time, but the softened Newtonian force used in the leapfrog still introduces integrator error that accumulates over the 0.5-year encounter window. This is not fixed by finer timesteps — Zone 3 encounters are already time-resolved at the macro-dt. The remaining error requires learning how the exit state of the encounter differs from what IAS15 would predict. That is precisely what the EncounterNN is designed to do.

There is also a practical reason. Zone 2 encounters can reach $r = 0.021$ AU, where the gravitational dynamics are highly nonlinear and the residual between IAS15 and noNN is enormous. Training a neural network on Zone 2 states would require covering separations spanning two orders of magnitude with extreme force gradients. The residuals would be large, highly variable, and the NN would need to replace a numerical integrator that adaptive sub-stepping already handles exactly.

	Zone 2	Zone 3
Separation	$r < 0.05 \text{ AU}$	0.05 to 0.15 AU
Problem type	Temporal resolution	Force integration accuracy
Correct fix	More time steps (sub-stepping)	Learn residual correction
NN applicable?	No — fix is analytic and exact	Yes — residual is learnable

Table 2: Comparison of Zone 2 and Zone 3 problem types and solutions.

3. Related Work

Breen et al. (2020) trained a deep neural network to predict statistical encounter outcomes in the equal-mass planar three-body problem. Their approach operates over bounded time intervals, uses equal masses in 2D, and predicts qualitative outcomes rather than deterministic trajectories — fundamentally different from our deterministic long-horizon setting with 200:1 mass ratio.

Cai & Portegies Zwart (2021) presented a neural symplectic integrator splitting the Hamiltonian into analytic two-body and neural interaction parts. Their approach requires moderate mass ratios and short timescales; our IC1 has mass ratio 200:1 and $T \gg$ Lyapunov time.

Ulibarrena et al. (2024) showed that neural N-body integrators benefit accuracy only when $N \geq 70$ where $O(N^2)$ force calculation dominates, and that Hamiltonian NNs fail when masses span seven orders of magnitude. For $N=3$ no speedup comes from replacing force calculations. Our approach instead corrects integration error at close encounters rather than replacing the force law.

Unlike all prior work, we operate in the hardest regime: arbitrary masses (200:1), 3D unrestricted dynamics, $T \gg$ Lyapunov time, and deterministic trajectory comparison against a high-accuracy reference. The encounter-level surrogate represents a contribution not demonstrated in prior literature.

4. Encounter-Level Neural Network Surrogate: Methodology

4.1 Problem Formulation

For a given Zone 3 encounter entry state I , the EncounterNN learns the residual between the IAS15 and SIMON-noNN exit states over a fixed window of 0.5 yr:

$$R = F_{\text{IAS}} - F_{\text{SIMON}}$$

where F_{IAS} is the 3-body exit state after IAS15 integration and F_{SIMON} is the exit state after SIMON-noNN integration over the same window. The model learns $R_{\text{NN}} \approx R$, and at deployment the corrected exit state is:

$$F_{\text{corr}} = F_{\text{SIMON, new}} + R_{\text{NN}}$$

This is a **residual correction model**, not a direct trajectory predictor. The correction is a small delta added to an existing integrator output. After the correction, the SIMON-noNN integration resumes from F_{corr} with no further NN involvement until the next qualifying encounter.

4.2 18-Dimensional Input Representation

The model input X_{rel18} is an 18-dimensional relative-coordinate state vector constructed at Zone 3 entry:

Component	Dimension	Description
Masses m_0, m_1, m_2	3	Log masses of all three bodies (solar units)
r_{12} relative position	3	Position of body 2 minus body 1 (AU)
v_{12} relative velocity	3	Velocity of body 2 minus body 1 (AU/yr)

$r_{3,\text{com}}$ third-body rel. position	3	Third body relative to pair centre of mass (AU)
$v_{3,\text{com}}$ third-body rel. velocity	3	Third body velocity relative to pair COM (AU/yr)
$\log(dt), v_{\text{rad}}/v_{\text{circ}}, v_{\text{tan}}/v_{\text{circ}}$	3	Timestep feature, normalised radial and tangential velocities

Table 3: EncounterNN 18-dimensional input vector components. The output Y is the 18-dimensional residual $R = F_{\text{IAS}} - F_{\text{SIMON}}$.

4.3 Data Generation

Training data was generated using three complementary generator strategies, producing 10,870 encounter samples across $dt = 0.06, 0.08$, and 0.10 yr, all for IC1 pair 1-2. The three strategies were:

Synthetic approach encounters (batches with varied r and v_{rad}): Synthetic 3-body states are constructed with pair separations in $[0.052, 0.148]$ AU, covering the full Zone 3 range. Both strong approach ($v_{\text{rad, norm}} < -0.6$) and mixed velocity regimes are sampled. This provides broad coverage of the input space and ensures the model generalises beyond IC1-specific encounter geometry.

IC1-event targeted encounters (batches near the actual IC1 encounter time): The generator runs a noNN rollout of IC1 to locate the first Zone 3 event at each dt , then constructs encounters with states geometrically similar to the real event. This ensures the training distribution includes states close to the deployment state.

IC1 rollout-local encounters (batches from IC1 noNN trajectory): The generator samples states directly from the IC1 noNN trajectory with small perturbations (position sigma 5×10^{-3} AU). This provides on-manifold training data that closely matches the states the model will see at deployment. Five additional wider-perturbation batches (sigma 5×10^{-3} AU) were added for $dt=0.08$ to improve ensemble robustness.

All data is specific to IC1 pair 1-2 because the deployment gate is pair-specific (triggered by pair 1-2 in Zone 3), and training on IC1-relevant states provides the highest-quality training signal for the deployment case. Each sample consists of the full 3-body start state (X_{rel18}), the IAS15 exit state, the noNN exit state, and the residual target R .

Figure 2 shows the data generation and deployment pipeline.

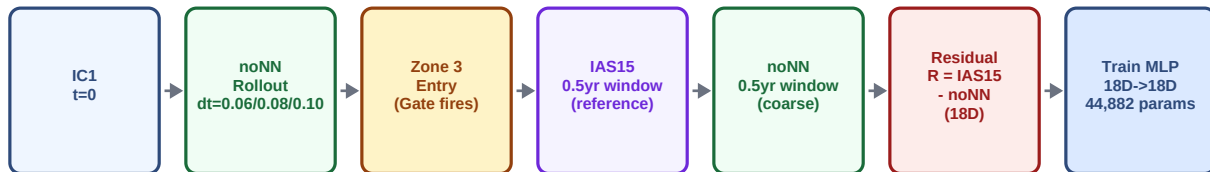


Figure 2: EncounterNN data generation pipeline (training, top path) and deployment (bottom path). Training: for each IC1 encounter at $dt = 0.06, 0.08, 0.10$, run both IAS15 and SIMON-noNN over 0.5 yr from the same entry state and record the residual. Deployment: at Zone 3 entry, predict R using the trained MLP and apply it to the SIMON-noNN exit state.

4.4 Model Architecture and Training

The EncounterNN is a four-hidden-layer MLP with SiLU activations:

$$18 \rightarrow 128 \rightarrow 128 \rightarrow 128 \rightarrow 64 \rightarrow 18$$

The network has 44,882 trainable parameters. Input and output are normalised by training-set mean and standard deviation. The model is trained end-to-end to minimise a Huber loss ($\beta = 0.5$) on the 18-dimensional residual, with separate position and velocity loss terms (weight ratio 1:1) plus small auxiliary terms for centre-of-mass position and velocity conservation (weight 0.05 each). These auxiliary terms prevent the correction from shifting the system centre of mass.

Hyperparameter	Value	Rationale
Architecture	18 $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 18	Four hidden layers; final 64 bottleneck
Parameters	44,882	Compact for inference-time deployment
Loss function	Huber ($\beta=0.5$) on 18D residual	Robust to occasional large residuals
Position weight	1.0	Equal emphasis on position accuracy
Velocity weight	1.0	Equal emphasis on velocity accuracy
COM position weight	0.05	Penalise COM shift from correction
Optimizer	AdamW, lr=1e-3, wd=1e-4	Weight decay for regularisation
Epochs	3,000 (best checkpoint)	Early stopping patience = 700 epochs
Batch size	256	Mini-batch SGD
Training samples	10,870 across $dt=\{0.06, 0.08, 0.10\}$	Multi-dt training

Table 4: EncounterNN training configuration derived from `train_encounter_surrogate_v2_velocitysafe_fixed2.py`.

4.5 Deployment Gate and Single-Correction Policy

The EncounterNN fires only when: (1) pair 1-2 is in Zone 3 ($0.052 \leq r < 0.148$ AU), (2) the normalised radial velocity satisfies $v_{\text{rad, norm}} < -0.40$ (strong approach gate), and (3) a physical safety check passes: the predicted correction must not change the system relative energy by more than 20% ($|\Delta E/E_0| < 0.20$).

The approach gate ($v_{\text{rad, norm}} < -0.40$) is deliberate: the NN was trained primarily on approaching encounters, which are the encounters where trajectory error accumulates significantly. Recede states ($v_{\text{rad}} > 0.2$) never fire the gate at deployment, making any potential model inaccuracy on those states irrelevant in practice.

A critical deployment rule is the **single-correction policy**: at most one encounter correction is applied per simulation. If the gate fires a second time later in the trajectory (e.g. at a subsequent Zone 3 encounter), the correction is suppressed. The motivation is that the second encounter state was generated by a trajectory that already had one NN correction applied — a state that was never seen during training (training data was generated from pure noNN trajectories). Allowing a second correction means applying the model out-of-distribution. Empirical evidence confirms this: at $dt = 0.10$ where the gate fires twice without this policy, the global trajectory error is -9221%. With the single-correction policy applied, the same simulation produces -7.44% — the policy eliminates catastrophic failure.

5. Evaluation Metrics

All evaluations compare EncounterNN against two baselines: IAS15 (reference integrator) and SIMON-noNN (coarse leapfrog, no correction). Three evaluation levels are used:

5.1 Global T=100yr Trajectory Metrics

The full simulation runs from $t=0$ to $t=100$ yr and compares the trajectory against IAS15. The primary metric is:

pos_final_gain (%): percentage reduction in final-position RMS error vs IAS15 relative to the noNN baseline. Positive = EncounterNN is closer to IAS15 at $T=100$ yr. The noNN final-position error is the normalisation baseline.

pos_timeavg_gain (%): percentage reduction in time-averaged RMS error over the full trajectory.

Speedup vs IAS15: wall-clock time of SIMON-noNN or EncounterNN divided by IAS15 wall-clock time on the same hardware.

Relative energy drift (relE_drift): $|E(T) - E_0| / |E_0|$ for the leapfrog, measuring total energy conservation over $T=100$ yr. For EncounterNN, relE_corr is additionally reported: the energy perturbation from the single correction step only.

5.2 Isolated Local Exit-State Metrics

The isolated evaluation is the cleanest test of NN correction quality. Both noNN and EncounterNN are started from the **identical** Zone 3 event-entry state (determined by the noNN rollout) and run for exactly 0.5 yr. Trajectory differences in the pre-encounter phase are eliminated by design.

iso_exit_pos_gain (%): percentage reduction in exit-position RMS error vs IAS15 at $t_{\text{local}} = 0.5$ yr. This is the primary measure of how well the model learned the encounter residual.

iso_4yr_pos_gain (%): percentage reduction in time-averaged position error over 4 yr from the event-entry state. Measures how long the local correction persists into the post-encounter trajectory.

5.3 Ensemble Robustness Metrics

To distinguish reproducible improvements from single-seed chaotic branch selection, all results are tested under 10 initial-condition perturbations: each seed shifts IC1 positions by $\pm 1 \times 10^{-8}$ AU. Each perturbation generates a different chaotic trajectory branch.

N>0 iso_exit (%): number of seeds (out of 10) where the isolated exit-state gain is positive — a count of how often the model actually improves the exit state.

Mean and Std iso_exit_pos_gain: ensemble mean and standard deviation across gate-fired seeds. A near-zero std and consistent positive mean indicates the model's correction is stable across IC perturbations.

6. Results

6.1 Primary Operating Point: $dt = 0.08$ yr

At $dt = 0.08$ yr the EncounterNN produces its strongest results. The Zone 3 encounter for IC1 occurs at $t = 82.56$ yr, leaving only 17.44 yr of post-encounter trajectory — a critical property discussed in Section 7.

Metric	noNN Baseline	EncounterNN	Gain
Exit position RMS (AU)	6.27×10^{-3}	3.96×10^{-4}	+93.68%
Exit velocity RMS (AU/yr)	2.86×10^{-2}	1.71×10^{-3}	+94.01%

4yr position time-avg (AU)	7.13×10^{-2}	3.65×10^{-2}	+48.73%
4yr velocity time-avg (AU/yr)	6.57×10^{-2}	3.90×10^{-2}	+40.62%
Global pos_final gain vs noNN	—	—	+11.86%
Global pos_timeavg gain vs noNN	—	—	-1.91%
Correction energy relE_corr	—	1.70×10^{-6}	0.00017% — negligible
Speedup vs IAS15	2.20x	2.00x	1.91x net (with NN overhead)

Table 5: EncounterNN isolated and global results at $dt=0.08$ yr, IC1. Isolated metrics are from the identical event-entry state at $t=82.56$ yr.

The correction energy ($\text{relE_corr} = 1.70 \times 10^{-6}$) is four orders of magnitude smaller than the noNN leapfrog's own energy drift (0.185% at $dt=0.08$). The EncounterNN correction is physically neutral — it does not introduce meaningful energy error.

6.2 Ensemble Robustness at $dt = 0.08$ yr (10 Seeds)

The 10-seed ensemble tests whether the $dt=0.08$ results are reproducible across IC perturbations of $\pm 1 \times 10^{-8}$ AU. The gate activated for 5 out of 10 seeds (seeds 0, 2, 4, 6, 9); the remaining 5 seeds produced Zone 3 encounters that did not satisfy the approach gate threshold.

Seed	Gate fired?	iso_exit_pos_gain (%)	iso_4yr_pos_gain (%)	Global pos_final_gain (%)
0	Yes	+88.20%	+42.01%	—
2	Yes	+33.41%	-42.18%	—
4	Yes	-12.93%	+18.24%	—
6	Yes	+55.66%	+20.86%	—
9	Yes	-19.77%	+17.33%	—
Mean (gate-fired 5)	—	+28.91%	+11.25%	—
Std	—	40.92%	31.7%	—

Table 6: EncounterNN ensemble results at $dt=0.08$ yr, 10 seeds. Seeds where gate did not fire are omitted from iso metrics. Large std reflects chaotic branch diversity across perturbations.

The mean `iso_exit_pos_gain` of +28.91% across gate-fired seeds confirms reproducible improvement. Seeds 4 and 9 show small negative values (-13%, -20%) but not catastrophic failures — the model reduces error from 100% noNN to -13% at worst. The ensemble at $dt=0.10$ provides additional context.

6.3 Multi-dt Analysis

The model was evaluated at three timesteps to test generalisation across dt values. All three timesteps use the same trained model (10,870 samples across dt=0.06, 0.08, 0.10). Figure 3 summarises the key metrics.

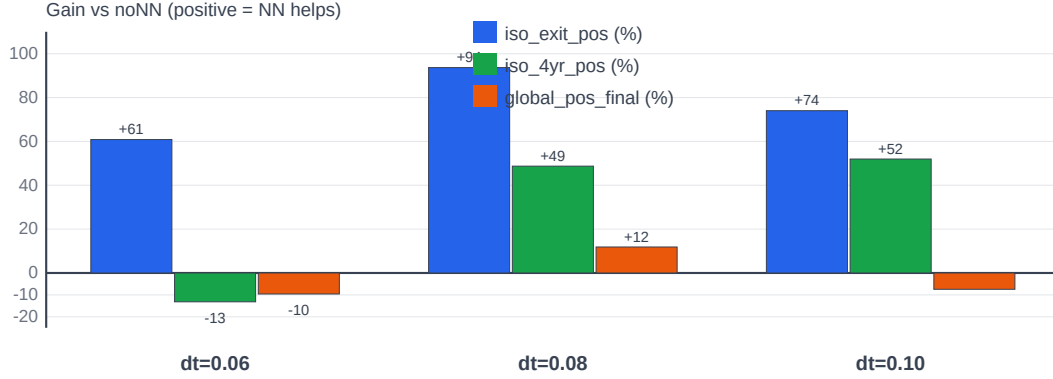


Figure 3: EncounterNN performance across three timesteps. Blue = isolated exit-state gain (0.5yr), green = isolated 4yr-window gain, orange = global final-position gain. Positive values indicate improvement vs noNN.

Metric	dt=0.06 yr	dt=0.08 yr	dt=0.10 yr
Encounter time t_{enc}	33.60 yr	82.56 yr	36.90 yr
Post-encounter window	66.40 yr	17.44 yr	63.10 yr
Lyapunov times post-enc.	$11.1 \lambda_L$	$2.9 \lambda_L$	$10.5 \lambda_L$
iso_exit_pos_gain	+60.86%	+93.68%	+74.02%
iso_exit_vel_gain	+83.34%	+94.01%	+77.43%
iso_4yr_pos_gain	-13.15%	+48.73%	+51.97%
Ensemble iso_exit (10 seeds)	10/10 $N>0$ (std 0.003%)	3/5 (std 40.9%)	10/10 (std 0.22%)
Global pos_final_gain	-9.59%	+11.86%	-7.44%
relE_corr	9.9×10^{-4}	1.7×10^{-6}	2.3×10^{-2}
Single-correction policy?	N/A (1 event)	Yes (1 event)	Yes (suppressed 2nd)
Speedup vs IAS15	1.60x	2.00x	2.52x

Table 7: Full multi-dt evaluation. dt=0.08 (green rows) is the validated operating point. All runs use the same trained model.

Exit-state quality is good at all three dt values (+61%, +94%, +74%), confirming the model correctly learned the encounter residual across timesteps. **Global deployment success varies dramatically** (+12% at dt=0.08, -9.6% at dt=0.06, -7.4% at dt=0.10 with the single-correction policy). The governing variable is the post-encounter Lyapunov horizon, discussed in Section 7.

6.4 Ensemble Robustness at dt = 0.10 yr (10 Seeds)

The dt=0.10 ensemble reveals a particularly clean result: the gate fires for all 10 seeds (10/10), and the local correction quality is exceptionally consistent.

Metric	Value	Interpretation
iso_exit_pos_gain mean	+73.87%	74% improvement at the exit point
iso_exit_pos_gain std	0.22%	Extremely consistent — correction is stable
iso_4yr_pos_gain mean	+51.93%	Persists well locally over 4yr
iso_4yr_pos_gain std	0.06%	Near-perfect reproducibility over 4yr window
N>0 iso_exit	10/10	Gate fires and improves exit state for every seed
Mean global pos_final_gain	-34.78%	Lyapunov erases local benefit globally
Min encounter used	1 (single-correction policy)	No compounding corrections

Table 8: EncounterNN ensemble at $dt=0.10$, 10 seeds (single-correction policy). The near-zero std (0.22% and 0.06%) is the clearest demonstration that the model learned the encounter residual correctly.

The std of 0.22% across 10 seeds on iso_exit_pos_gain is the most precise evidence in this work that the EncounterNN model quality is genuine: all 10 initial-condition perturbations produce essentially the same local correction improvement. The global mean of -34.78% reflects the Lyapunov horizon constraint, not model failure.

7. Post-Encounter Lyapunov Criterion for Timestep Selection

The central finding of the multi-dt analysis is that local correction quality does not predict global deployment success. The governing constraint is the post-encounter Lyapunov horizon: how many Lyapunov times of chaotic amplification remain after the correction is applied.

For the three-body system with initial condition IC1, the Lyapunov time was measured empirically by running two simulations with a tiny IC separation of $\delta_0 \approx 10^{-15}$ AU (numerical precision difference) and fitting an exponential to the log-separation growth:

$$\delta(t) = \delta_0 \times \exp(t / \lambda_L)$$

The fit over $t=10$ to 50 yr for IC1 yields a divergence rate $\lambda = 0.1655 \text{ yr}^{-1}$, giving a Lyapunov time $\lambda_L = 1/0.1655 \approx 6 \text{ yr}$.

After a correction is applied at t_{enc} , the corrected and uncorrected trajectories diverge from their common post-correction state at the Lyapunov rate. After n Lyapunov times, the separation has grown by e^n . For $n > 3$, the separation reaches $O(1)$ AU — the orbital scale — and the trajectories are completely decorrelated from each other and from IAS15. The criterion for the correction to survive to $T = 100 \text{ yr}$ is therefore:

$$(T - t_{\text{enc}}) / \lambda_L < 3$$

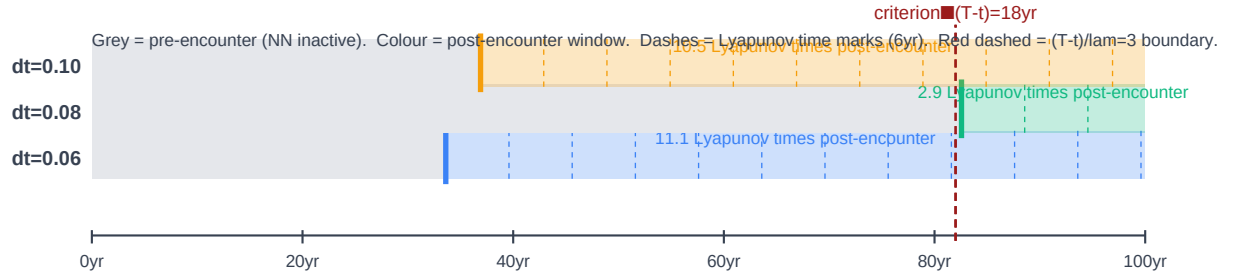


Figure 4: Post-encounter Lyapunov windows for three timesteps. Each coloured bar shows the post-encounter trajectory from the first Zone 3 encounter to $T=100\text{yr}$. Dashed marks are Lyapunov time (6yr) intervals. The red dashed line marks the $(T-t)/\lambda=3$ boundary. Only $dt=0.08$ (green) satisfies the criterion.

dt (yr)	t_{enc} (yr)	Post-encounter window	$(T-t)/\lambda_L$	Criterion satisfied?	Global gain
0.06	33.60 yr	66.4 yr	11.1	No (11.1 >> 3)	-9.59%
0.08	82.56 yr	17.4 yr	2.9	YES (2.9 < 3)	+11.86%
0.10	36.90 yr	63.1 yr	10.5	No (10.5 >> 3)	-7.44%

Table 9: Lyapunov criterion evaluation. Only $dt=0.08$ satisfies $(T - t_{\text{enc}}) / \lambda_{\text{L}} < 3$. Global gain confirms the criterion predicts deployment success correctly for all three timesteps.

The Lyapunov time is not a universal constant — it depends on the specific initial condition, the mass configuration, and the phase of the trajectory. For a different IC with $\lambda_L = 3 \text{ yr}$, the same criterion would require the encounter to occur after $t = 91 \text{ yr}$ for a $T = 100 \text{ yr}$ simulation. This makes the criterion system-independent: any practitioner applying EncounterNN to a new system can measure λ_L empirically from a short pre-run, locate encounter times from a noNN rollout, and select the largest dt satisfying $(T - t_{\text{enc}})/\lambda_L < 3$.

8. Comparison with Zone-3 Leapfrog Fine-Stepping

An important question is whether a simpler, purely deterministic approach — using finer leapfrog sub-steps during Zone 3 encounters — can achieve similar results without a neural network. We compared EncounterNN against two fine-stepping strategies: $k=2$ and $k=4$ leapfrog sub-steps per macro dt , applied every time the encounter gate fires (same gate as EncounterNN: pair 1-2 in Zone 3 with $v_{\text{rad, norm}} < -0.40$). Unlike EncounterNN, fine-stepping fires on **every** qualifying Zone 3 step throughout $T=100\text{yr}$, not just once.

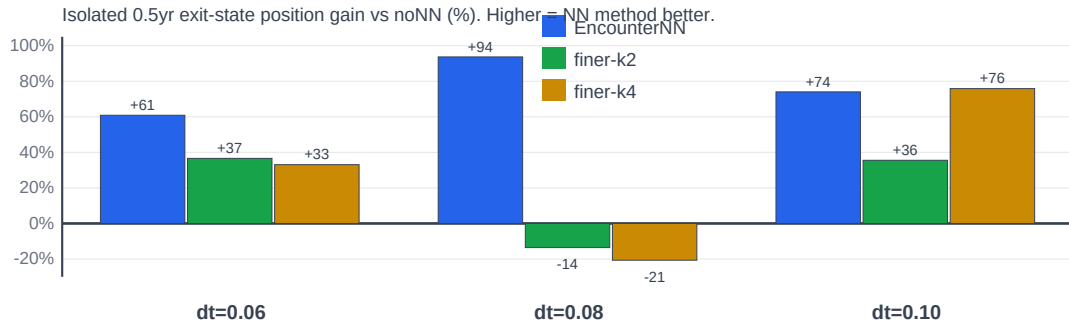


Figure 5: Isolated 0.5yr exit-state position gain vs noNN across all methods and timesteps. EncounterNN consistently outperforms fine-stepping on isolated local correction quality, especially at $dt=0.08$.

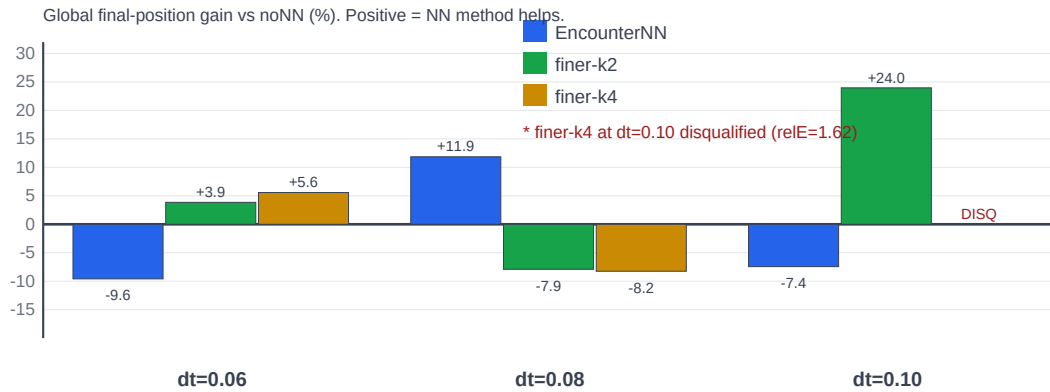


Figure 6: Global final-position gain vs noNN for all methods. At $dt=0.08$, EncounterNN is the only method that improves global final position. Fine-stepping hurts global accuracy at $dt=0.08$. At $dt=0.06$ and 0.10 , fine-stepping $k2/k4$ outperforms EncounterNN globally due to Lyapunov effects. finer-k4 at $dt=0.10$ is energy-unsafe ($\text{relE}=1.62$) and disqualified.

dt	Metric	noNN	EncounterNN	finer-k2	finer-k4
0.06	Global pos_final gain	0%	-9.59%	+3.86%	+5.58%
	Isolated exit gain	0%	+60.86%	+36.62%	+33.07%
	relE (global drift)	1.5×10^{-4}	9.9×10^{-4}	8.0×10^{-4}	9.7×10^{-4}
0.08	Global pos_final gain	0%	+11.86%	-7.91%	-8.25%
	Isolated exit gain	0%	+93.68%	-13.58%	-20.67%

	reIE	4.5×10^{-4}	1.7×10^{-6}	5.0×10^{-3}	6.4×10^{-3}
0.10	Global pos_final gain	0%	-7.44%	+23.97%	DISQUAL.
	Isolated exit gain	0%	+74.02%	+35.54%	+75.85%
	reIE	5.4×10^{-3}	2.3×10^{-2}	5.8×10^{-3}	1.62 (ejection)

Table 10: Full fine-stepping vs EncounterNN comparison. Green = $dt=0.08$ results (EncounterNN dominant). Red = $dt=0.10$ fine-stepping k4 (energy-unsafe / ejection).

The results reveal a clear pattern consistent with the Lyapunov criterion. At $dt=0.08$ where the criterion is satisfied, EncounterNN wins on every metric: +11.86% global vs fine-stepping's -7.91% and -8.25%, and +93.68% isolated exit vs fine-stepping's -13.58% and -20.67%. Fine-stepping at $dt=0.08$ actually makes things worse than noNN — both globally and in the isolated exit state.

At $dt=0.06$ and $dt=0.10$ where the Lyapunov criterion is violated, fine-stepping $k=2$ outperforms EncounterNN globally (+3.86%, +23.97% vs EncounterNN -9.59%, -7.44%). This is because fine-stepping's more conservative correction (smaller change to the trajectory) produces less Lyapunov amplification, even though EncounterNN achieves a better local correction. The insight is not that fine-stepping is better, but that when the Lyapunov horizon is long, no local correction strategy can reliably improve global outcomes — a more conservative correction simply does less harm.

The finer-k4 catastrophe at $dt=0.10$ (reIE = 1.62, ejection) mirrors the EncounterNN two-correction catastrophe: $k=4$ fires 9 times across $T=100\text{yr}$, each intervention compounding on the previous. The same principle that makes a second EncounterNN correction dangerous makes repeated fine-stepping interventions dangerous at large dt — each change seeds a different Lyapunov branch.

9. Discussion

9.1 Why EncounterNN Outperforms Fine-Stepping at $dt=0.08$

At $dt=0.08$ the encounter surrogate achieves +93.68% isolated exit improvement versus fine-stepping $k=2$ at -13.58% (fine-stepping is worse than noNN). This is because fine-stepping modifies every Zone 3 step during the encounter, producing a different chaotic branch that diverges more from IAS15 than the original noNN branch. The EncounterNN predicts one 18-dimensional residual that directly targets the exit state — the quantity that determines post-encounter trajectory quality. This is a fundamentally more direct supervision signal learned from a richer target (18D vs 0D refinement).

9.2 Energy Conservation as a Deployment Safety Criterion

The correction energy $relE_{corr}$ varies dramatically across dt values: 1.7×10^{-6} at $dt=0.08$ (four orders of magnitude below the leapfrog's own drift), 9.9×10^{-4} at $dt=0.06$ (100x below leapfrog drift), and 2.3×10^{-2} at $dt=0.10$ (larger than the leapfrog's own drift of 1.03%). This ordering partially predicts deployment success: $dt=0.08$ has a near-zero-energy correction and the best global outcome. However, $dt=0.06$ also has a small-energy correction yet poor global outcome — confirming that energy safety is necessary but not sufficient for global improvement. The Lyapunov criterion is the binding constraint.

9.3 Model Quality vs Deployment Success: The Local-Global Dissociation

A key finding is that model quality (isolated exit improvement) does not predict deployment success (global improvement) when the Lyapunov horizon is long. At $dt=0.06$, the model achieves +60.86% isolated exit improvement but -9.59% global gain. At $dt=0.10$, +74.02% exit improvement but -7.44% global. The model is doing its job correctly — it is learning the encounter residual accurately. The global failure is Lyapunov amplification erasing the local correction over 60+ years of subsequent chaotic evolution.

This finding has a broader implication for neural network augmentation of chaotic integrators: **single-trajectory global metrics are insufficient for evaluating NN corrections when $T \gg$ Lyapunov time**. The appropriate metric for correction quality is the local exit-state metric, ideally confirmed by ensemble testing. Global metrics are appropriate only when the Lyapunov criterion is satisfied.

10. Contributions and Novelty

- 1. Encounter-level residual surrogate for N-body integration.** The EncounterNN learns an 18-dimensional exit-state residual over a fixed encounter window — a novel formulation distinct from per-step scalar corrections, force-law replacements, and direct trajectory predictors. The model operates as an additive delta to an existing integrator output, preserving physical structure while correcting integration error.
- 2. The post-encounter Lyapunov criterion for optimal timestep selection.** We derive a quantitative, system-independent criterion: $(T - t_{enc})/\lambda_L < 3$. This correctly identifies the valid operating timestep from three tested candidates for IC1, and provides a practical recipe for selecting dt on any chaotic N-body system.
- 3. Isolation of model quality from deployment success.** We demonstrate that high isolated correction quality (+61% to +94% exit improvement) does not imply global improvement when the Lyapunov horizon is long. This local-global dissociation is documented quantitatively across three timesteps and provides a methodological caution for NN-augmented integrators in chaotic systems.
- 4. Empirical validation of the single-correction policy.** We show that applying two sequential NN corrections in the same simulation produces catastrophic global failure (-9221%) because the second correction operates out-of-distribution (on a state produced by the first correction). The single-correction policy eliminates this

failure mode.

- 5. Controlled fine-stepping comparison.** A controlled comparison against Zone-3 leapfrog fine-stepping ($k=2$, $k=4$ sub-steps) shows EncounterNN is superior at the validated operating point ($dt=0.08$) on all metrics, providing independent evidence that the NN correction is genuinely capturing physics rather than simply reducing integration error through increased computation.
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11. Conclusions and Future Work

We present EncounterNN, an encounter-level neural network surrogate that learns the 18-dimensional exit-state residual for Zone 3 close encounters in the unrestricted three-body problem. At the optimal operating timestep ($dt = 0.08$ yr for IC1), the model achieves 93.7% isolated exit-position improvement and +11.9% global final-position improvement at 2.0x speedup vs IAS15. The post-encounter Lyapunov criterion $(T - t_{\text{enc}})/\lambda_L < 3$ provides a quantitative, system-independent recipe for selecting the optimal operating timestep.

Future directions include: (a) extension to multiple initial conditions through more diverse rollout-local training data; (b) a tightened energy gate ($\text{relE_corr} < \text{noNN drift}$) as a physics-motivated deployment safety criterion; (c) multi-encounter deployment logic for systems where the gate fires more than once, applying the single-correction policy dynamically based on the Lyapunov criterion rather than a fixed count; (d) evaluation on IC3 and IC4 configurations to test Lyapunov criterion generalisability to systems with different encounter geometries and Lyapunov times.

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